

Stress-Testing a Provisional Analysis for the Biomarker-by-Drug Interaction in the Hybrid Design: Pre-Registration

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Purpose. This document fixes, before any of the simulations it describes are run, how the project will choose its analysis for the biomarker-by-drug interaction in the Hybrid design.

It names a *strawman*: a provisional analysis, put forward to be attacked. It defines four criteria the analysis must meet: nominal, robust, powerful and sensitive. It lists the challengers, and a set of stress conditions designed to break the strawman. It then states the rule by which the strawman survives or is replaced.

The strawman is chosen on the evidence in docs/45, read under the strict size criterion of Section 3. It is not the primary analysis recommended there. Section 8 explains the change.

The study follows the ADEMP structure: aims, data-generating mechanisms, estimands, methods, performance measures (Morris, White and Crowther [1]).

Notation index and glossary

Notation follows analysis/report/NOTATION.md and docs/45. Symbols marked *local* are defined here.

Symbol	Meaning	Status
Y_{it}	symptom score (code Sx); $t = 0$ is baseline	canonical
$BR_{it}, PB_{it}, TV_{it}$	drug, expectancy and natural-course responses	canonical
B_i, b_i	biomarker; standardized biomarker	canonical
$D_{it}, D_{bc,it}$	binary drug state; exposure-decayed indicator, 0 before any exposure	canonical
$t_{1/2}, t_{1/2}^a$	true carryover half-life; half-life assumed by the analysis	canonical; local (docs/45)
c_{bm}	biomarker-BR coupling (the predictive effect)	canonical

Symbol	Meaning	Status
$c_{bm,TV}, c_{bm,PB}$	biomarker-TV and biomarker-PB correlations (prognostic couplings)	local; $c_{bm,PB}$ is the package's <code>c.bm.pb</code>
β_{bm}^{TV}	mean moderation of the natural course by b_i	canonical (component slope)
$\beta_B, \beta_{bm:D}$	biomarker main effect; the interaction, the estimand	local (docs/41); canonical
ρ	AR(1) serial correlation per week	canonical
$\sigma_{\varepsilon,OL}, \sigma_{\varepsilon,bl}$	SD of added measurement error, open-label and blinded visits	local
κ	mean model SE over empirical SD	canonical
π	power	canonical
N	total sample size	canonical

Glossary.

- **Strawman.** A provisional analysis put forward to be criticized and tested, not a deliberately weak one. In docs/36 and docs/44 the word named the published-code reference arm; that arm is called the *published reference arm* here.
- **Challenger.** An analysis that could replace the strawman.
- **Stress cell.** A condition chosen because it is expected to break at least one candidate.
- **Power curve.** Size-adjusted power as a function of the true coupling c_{bm} , the basis of the sensitivity criterion.
- **Base cell.** The condition from which each stress varies one factor.
- **Prognostic biomarker.** A biomarker associated with the outcome regardless of treatment, here through TV or PB.

1. Aims

1. Test whether the strawman (Section 4) holds its nominal size in every stress cell.
2. Compare its power and sensitivity with every challenger that also holds its size.
3. Replace the strawman if, and only if, the rule of Section 7 says so.
4. Report the published reference arm alongside, for comparability with Hendrickson et al. [2].

2. Scope

- **Design:** Hybrid only. Visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, plus baseline. Four paths with the published 18/18/17/17 allocation at $N = 70$, scaled proportionally at other N .
- **Estimand:** $\beta_{bm:D}$.
- **Test:** two-sided at $\alpha = 0.05$.

3. Criteria (fixed before the run)

Nominal. Two conditions must both hold:

- the pooled rejection rate over all null cells lies within 2 Monte Carlo SE of 0.05;
- no null cell exceeds 0.075.

Each null cell has 1,000 replicates, so the MCSE is 0.0069 per cell. There is no “slightly liberal” class. An analysis that fails either condition fails the criterion.

Robust. Nominal in every stress cell of Section 5, each assessed separately with the cell condition above. A failure in a single stress cell is recorded with the condition that caused it.

Powerful. Power is size-adjusted: each analysis rejects at the 5% quantile of its own p-values in the matching null cell. The summary is the maximin shortfall. In each alternative cell, compute each analysis’s shortfall from the best nominal analysis in that cell. The summary for an analysis is its largest shortfall over the cells. Unadjusted power is also reported.

Sensitive. The test should respond to the strength of the true interaction: power should rise steadily as the true coupling c_{bm} moves away from zero in either direction, and reach α at zero. It is assessed on the power curve, size-adjusted, over the grid of Section 5. Its components are:

- **monotonicity:** no decrease in power between adjacent grid points larger than 2 MCSE, on either side of zero;
- **symmetry:** power at $-c_{bm}$ within 2 MCSE of power at $+c_{bm}$;
- **steepness:** the average slope of power against $|c_{bm}|$ between 0.15 and 0.30;
- **detectable coupling:** the smallest $|c_{bm}|$ at which power reaches 0.80, by linear interpolation on the grid. Smaller is more sensitive.

The curve is estimated in two constructs. In the base construct (AR(1)) $|c_{bm}|$ runs to 0.45. In the compound-symmetry construct it runs to 0.25, the largest value below that construct’s ceiling of 0.256.

Estimand fidelity is also reported, though it is not a selection criterion: the relative bias of $\hat{\beta}_{bm:D}$ against the value the same analysis gives at $t_{1/2} = 0$, when $t_{1/2}^a$ is close to $t_{1/2}$.

4. The strawman and the challengers

Strawman S. Random intercept plus `corCAR1` residuals, with CR2 standard errors and Satterthwaite degrees of freedom (Bell and McCaffrey [3]). Baseline is a response row, coded off drug. The model uses linear time and $D_{bc,it}$ at $t_{1/2}^a = 0.5$ week:

```
nlme::lme(Sx ~ bmc + t + Dbc + bmc:Dbc, random = ~ 1 | ptID,
           correlation = nlme::corCAR1(form = ~ t | ptID))
# then clubSandwich::coef_test(fit, vcov = 'CR2',
#                               test = 'Satterthwaite')
```

It is chosen because it is the most powerful analysis that was nominal, under the strict criterion, in all eight null cells of `docs/45`. Its sizes there were 0.040 to 0.064, with a mean of 0.052.

Challengers. All are fitted to the same trials.

ID	Analysis	Reason
A	unstructured (<code>us</code>), Kenward-Roger, baseline row, linear t , $D_{bc,it}$ at 0.5	primary of <code>docs/45</code> ; most powerful under AR(1) there
C1	S with <code>varIdent(form = ~ 1 \ phase)</code> , phase = baseline / open label / blinded; CR2	gives the parsimonious model the phase heteroscedasticity that likely drives A's advantage
C2	C1 with model-based (Satterthwaite) inference	whether CR2 costs power once the variance model is closer
DS	S with $t_{1/2}^a$ chosen by AIC over $\{0, 0.25, 0.5,$ $1, 2\}$ (0 meaning D_{it}); ML for selection, REML refit	data-chosen half-life
DA	A with the same AIC rule	the same, for the unstructured model

ID	Analysis	Reason
E	the best of S, A, C1, DS and DA (by the base-cell results), with β_B freed at baseline (+ bmc:b10)	robustness to failure of the constancy assumption
F1	S plus bmc:De , where De is the expectancy (1 open label, 0.5 blinded)	remedy for a placebo-prognostic biomarker
F2	S plus bmc:t	remedy for a natural-course-prognostic biomarker
R	published reference arm: random intercept, model-based, linear t , binary D_{it} (cs-Satt in docs/45)	comparability

E, F1 and F2 change the model, not only the inference. They are evaluated in every cell, so their cost when nothing is wrong is measured as well as their benefit when something is.

5. Data-generating mechanisms

Base construct. Configuration C of **docs/36**:

- separable AR(1) within each component at $\rho = 0.7$ per week;
- graded coupling of the biomarker to BR;
- published Gompertz means and SDs;
- the biomarker coupled only to BR.

This is the construct **docs/44** identifies as the best implemented. The base cell is:

- $N = 70$, $t_{1/2} = 0.5$ week, no dropout, exponential decay.

Power-curve cells (the sensitivity criterion):

Construct	Grid of c_{bm}
base (AR(1), $\rho = 0.7$)	$-0.30, 0, 0.075, 0.15, 0.225, 0.30, 0.375, 0.45$

Construct	Grid of c_{bm}
compound symmetry, 0.8 at every lag	-0.20, 0, 0.05, 0.10, 0.15, 0.20, 0.25

The zero cells have 1,000 replicates, all others 500. A negative c_{bm} means the biomarker is coupled to a smaller drug response. Under graded coupling the joint distribution exists for negative values on the same terms as for positive ones, which is checked before the run. The base-construct cells at 0 and 0.30 double as the null and alternative of the base cell.

Stress cells. Each varies one factor from the base cell. Each has a null twin ($c_{bm} = 0$, 1,000 replicates) and an alternative at $c_{bm} = 0.30$ (500 replicates), unless the table says otherwise.

#	Factor	Level	Expected to break
1	correlation	compound symmetry, 0.8 at every lag (docs/45 CS)	pure-AR(1) assumptions; tests S under its weakest structure
2	correlation	base plus measurement error, $\sigma_{\varepsilon,OL} = 3$, $\sigma_{\varepsilon,bl} = 6$	homoscedastic working models (S, F1, F2)
3	sample size	$N = 35$	us with Kenward-Roger (A, DA); CR2 degrees of freedom
4	sample size	$N = 150$	none expected; shows whether any excess persists with N
5	carryover	$t_{1/2} = 0$	$D_{bc,it}$ when there is no carryover
6	carryover	$t_{1/2} = 1$ week	binary D_{it} (R); short $t_{1/2}^a$
7	decay form	Weibull, $k = 0.5$ (heavy tail)	the exponential $D_{bc,it}$
8	decay form	Weibull, $k = 2$ (light tail)	the exponential $D_{bc,it}$
9	dropout	20% monotone, completely at random	covariance estimation (A)

#	Factor	Level	Expected to break
10	dropout	20% monotone, at random given the last observed Y	all; MAR is assumed by every likelihood analysis
11	prognostic	$c_{bm,TV} = 0.3$ (time-constant natural-course association)	the shared β_B of the row coding (S, A, C)
12	prognostic	$\beta_{bm}^{TV} = 0.3$ (association growing along the TV curve)	every analysis without <code>bmc:t</code>
13	prognostic	$c_{bm,PB} = 0.3$ (expectancy-scaled placebo association)	every analysis without <code>bmc:De</code>

For the prognostic cells 11 to 13, the alternative twin keeps the prognostic coupling as well. The question is whether the remedies recover the interaction, not only whether the null holds.

Implementation of the new mechanisms in the `04-power-simulation.R` construct. The biomarker is index 1 of the joint vector.

- $c_{bm,TV}$, $c_{bm,PB}$. Set the biomarker's correlation with every post-baseline TV (or PB) element to the stated value. The PB SD is already scaled by the expectancy e_t , so the implied slope is expectancy-scaled. If the matrix is not positive definite the cell stops with an error; no repair is permitted.
- β_{bm}^{TV} . After the draw, add $\beta_{bm}^{TV} b_i \sigma_{TV} G_{TV}(t) / \max_t G_{TV}(t)$ to TV_{it} . This is mean moderation that grows along the natural-course curve.
- **Measurement error.** Add independent normal error to Y_{it} , with SD 3 at open-label visits and 6 at blinded visits; none at baseline.
- **Weibull decay.** Applied to both the BR carryover and the graded coupling, as in manuscript 02's decay family. The analysis keeps the exponential $D_{bc,it}$.
- **Dropout.** Monotone, starting at a post-baseline visit:
 - MCAR: drawn uniformly over visits 3 to 8 for 20% of participants;
 - MAR: the hazard at each visit increases with the previous observed $Y_{i,t-1}$ (logistic, calibrated to 20% overall).

6. Methods and performance measures

Replicates: 1,000 per null cell and 500 per alternative cell, with per-chunk seeds as in `08-covariance-study.R`. All analyses are fitted to the same trials.

Reported for each analysis and cell:

- rejection rate with its MCSE;
- size-adjusted power;
- mean estimate, empirical SD, mean model SE and κ ;
- relative bias where defined;
- the distribution of the selected half-life (DS, DA);
- failed fits.

Failed fits count as non-rejections, and their rate is reported. An analysis with more than 2% failed fits in any cell is flagged.

Paired comparisons. Differences in power between two analyses in a cell are tested with McNemar’s test on the paired rejections.

7. Decision rule

1. **Screen.** Discard every analysis that fails the nominal criterion in any null cell, base or stress, except the prognostic stress cells 11 to 13.
2. **Prognostic screen.** Cells 11 to 13 are judged separately. An analysis that fails one of them is not discarded. It is flagged, and recommended only together with the remedy (E, F1 or F2) that is nominal in that cell.
3. **Sensitivity screen.** Flag any survivor whose power curve fails the monotonicity or symmetry condition in either construct; a flagged analysis may not be selected.
4. **Select.** Among the remaining survivors, select the analysis with the smallest maximin shortfall over the alternative cells, power-curve cells included. Report the detectable coupling and steepness of every survivor, and break ties within 0.02 of maximin shortfall by the smaller detectable coupling in the base construct.
5. **Strawman rule.** S is retained unless a survivor beats it on the maximin shortfall by more than 0.05, with the McNemar difference significant at 0.05 in at least half the alternative cells. Ties go to the analysis with fewer covariance parameters.
6. **Replacement.** If S is replaced, the winner is reported as the recommended analysis. It is not stress-tested again in this study, because doing so would make the selection adaptive.

These rules are fixed by this document. Any change after results are seen will be reported as a deviation, with its reason.

8. Relation to earlier documents

- **docs/45 recommended A as primary.** Under the strict criterion of Section 3, A’s mean size of 0.058 fails the nominal criterion (the 2-MCSE band is 0.043 to 0.057). The “slightly liberal” class of docs/45 was defined after the results were seen. This document removes it, and S becomes the strawman.
- **The parsimonious challengers are given the feature A exploited.** In docs/45 they were not. C1 adds the phase-specific variance that was the likely source of A’s power advantage.
- **The constancy assumption is now tested.** docs/45 could not test it, because every construct satisfied it. Cells 11 to 13 do.

9. Compute and execution

Size. 15 power-curve cells (8 base, 7 compound symmetry) plus 13 stress cells $\times 2$, about 40 cells once the curve cells shared with the stress set are counted once. About 12 analyses per replicate, with about 24,000 replicates in total. A, DA and E-on-A use unstructured Kenward-Roger fits and dominate the cost.

Time. About four to five days on 8 cores, scaled from the exposure-coding pass of docs/45 (inferred). The run should go to the simulation server if available.

Order of work.

1. Implement the mechanisms of Section 5 and the analyses of Section 4 in a new driver, `11-strawman-stress-test.R`. Add unit checks:
 - the positive-definiteness check on every new matrix;
 - the prognostic couplings reproduce their target correlations in a large draw;
 - the dropout rates match their targets;
 - the AIC rule reproduces the fixed $t_{1/2}^a$ analysis when it selects it.
2. A pilot of 20 replicates per cell, to check every analysis fits and to time the run.
3. The full run.
4. A results document (docs/47), reporting against this pre-registration, deviations included.

10. Limitations known in advance

- One design (Hybrid) and one construct family (Gompertz components with separable AR(1) or CS).
- The prognostic couplings and the measurement-error SDs are chosen values, not estimates from trial data.

- Stress factors are varied one at a time, so interactions between stresses (for example small N with dropout) are not tested.
- The strawman’s own analysis half-life (0.5 week) and the AIC grid are fixed here. Other values are not explored.

11. References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine* 2019; 38(11):2074-2102. doi:10.1002/sim.8086.
2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013.
3. Bell RM, McCaffrey DF. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology* 2002; 28(2):169-181.
4. pmsimstats team. docs/36, docs/37, docs/41, docs/44, docs/45.